

Course Syllabus

General Information

Faculty	Information Technology		
Department	Software Engineering		
Semester	Second	Academic Year	2025/2026

Course Details

Course Title	Secure Software Development			Course No.	0444503		
Type of Learning	☒ On- campus ☐ Blended (..... Platform) ☐ Online (..... Platform)						
Credit Hours	3	Theoretical	3	Practical	0		
Course Level (according to JNQF)			6	JNQF Hours*			
Pre-requisite	0413201 – بناء تطبيقات الويب			Co-requisite			

Course Instructor Information

Instructor Name	Madallah Al-Majali		
Instructor E-mail	malmajali@meu.edu.jo		
Course Time	14:00- 15:00 (Sun, Tue, Thr)		
Office Hours	Mon(12:30-14:00)		
Office Number	N346	Office Phone	
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

Security-Driven Software Development, By **Aspen Olmsted**, Packt Publishing, **1st Edition**, 2024

2. References

1. Beginning Software Engineering, Rod Stephens, ISBN: 978-1-118-96914-4, 2015
2. Bernd Bruegge and Allen Dutoit (2013): Object-Oriented Software Engineering Using UML, Patterns, and Java (3rd ed.), Pearson Publishing, ISBN-13: 978-0136061250.
3. A. Puntambekar (2020): Object Oriented Analysis and Design for BE Anna University R17 CBCS (V-CSE - CS8592) , 2nd ed. Technical Publications. ASIN: B0892FPB68

Teaching Methods

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|----------------|---------------------------|-------------------------------|
| 1. Lectures | 2. Case Studies | 3. Interactive demonstrations |
| 4. Assignments | 5. Discussion & Questions | |

Course Description

Introduction to the principles and practices of **secure software development** provides students with the knowledge and practical skills necessary to design and implement secure software systems. It focuses on integrating security principles throughout the Software Development Life Cycle (SDLC), emphasizing threat modeling, secure design, and coding practices to prevent common vulnerabilities such as SQL injection, XSS, and insecure authentication. Students will explore compliance and privacy requirements, cryptography, and security testing techniques (SAST, DAST) to ensure software resilience against attacks. Through case studies and project-based learning, students will gain hands-on experience in applying secure development frameworks and best practices to real-world applications.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Illustrate fundamental concepts and principles of secure software development and their importance in modern software systems.	PLO2		S1-D	
2	Identify and define software security requirements, including regulatory, compliance, and privacy aspects.	PLO2		S1-I	
3	Display secure design principles and threat modeling techniques (e.g., STRIDE, DREAD, CVSS) to analyze and mitigate security risks.	PLO2		S1-P	
4	Recreate secure coding practices to protect applications from common vulnerabilities such as XSS, SQL injection, weak authentication, and insecure session management.	PLO6		S2-P	
5	Follow cryptography and error-handling mechanisms to ensure data confidentiality, integrity, and reliability within software systems.	PLO6		S2-I	
6	Display best practices for maintaining and operating secure software systems, ensuring alignment with organizational policies and quality standards.	PLO6		S2-P	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture Type	Lecture No.		Assessment Tool	Summative/ Formative
1	*Security Software Concepts	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1	Discussion & Questions	Formative
2	*Understand Core Concept	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1	Discussion & Questions	Formative
3	*Understand Security Design Principles	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1	Discussion & Questions	Summative
4	*Secure Software Requirements	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	2	Quizzes	Summative
5	*Define Software Security Requirements *Identify Compliance Requirements	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	2	Discussion & Questions	Formative
6	*Identify Data Classification Requirements *Identify Privacy Requirements	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	2	Discussion & Questions	Summative
7	*Secure Software Architecture and Design *Threat Modeling *STRIDE *DREAD *CVSS	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
8	Mid Exam					Summative
9	*Secure Software Implementation *HTTP Request and Response	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
10	*Secure Coding practices for	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative

	Input Validation and Sanitization *XSS *SQL Injection	Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Summative
11	*Secure Coding practices for Cryptography	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	4	Quizzes	Summative
12	*Secure Coding practices for Authentication And Authorization	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	4	Discussion & Questions	Formative
13	*Secure Coding practices for Session Management	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	4	Discussion & Questions	Formative
14	*Secure Coding practices for Error Handling *Static and Dynamic Application Security Testing (SAST & DAST)	Lecture (1)	Lecture On- campus	5	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	5	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	5	Discussion & Questions	Formative
15	Project Research Discussion	Lecture (1)	Lecture On- campus	6	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	6	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	6	Discussion & Questions	Formative
16	Final Exam	Lecture (1)	Lecture On- campus	1-6		Summative

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Mid Exam	25%	Week 8
Quizzes	20%	Week 4, 11
Project	15%	Week 15
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	- No makeup quizzes. - If more quizzes are given, then the lowest quiz's grade is dropped.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	The last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exams or quizzes is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. - Sign-in sheets will be circulated. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Assignments are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (If Applicable)

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* JNQF Hours Calculation

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3	13.5	40.5
Tutorials	4	5	20
laboratory	0	0	0
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)			
Final Exam	2	1	2
Quizzes	1/2	2	1
Self-Learning Activities			
Assignments/ Homework	3	5	15
Case Studies	3	1	3
Presentations	3	1	3
E-learning			
Extra Readings	3	12	36
Total Hours	121.5		
JNQF Hours	Total Hours/10 = 12.15		